MTH313M Quiz Hints (April 7, 2026)

Question 1. (4 marks) Let $X_t = \sum_{j=1}^{N_t} Y_j, t \geq 0$ be a Compound Poisson Process with rate $\lambda > 0$ and component distribution F (i.e. $Y_1, Y_2, \cdots$ are i.i.d. with common DF F). Compute $Cov(X_s, X_t)$ for $0 \leq s \leq t < \infty$

Hints: Due to independent increments property,

$$Cov(X_s, X_t) = Cov(X_s, X_t - X_s + X_s) = Cov(X_s, X_s) = Var(X_s)$$

Question 2. (6 marks) Suppose that shocks occur in a system according to a Poisson Process with rate $\lambda > 0$. Further assume that the shocks indepedently causes the system to fail with probability $p \in (0,1)$ and that the system continues to work till it fails. Let M denote the number of shocks finally causing the system to fail and T denote the time of failure. Compute $\mathbb{P}(M = m | T = t)$ for $m = 1, 2, \cdots, t \in (0, \infty)$.

Hints: The number of shocks required for the system to fail follows Geometric distribution.

Question 3. (3 + 2 marks) Suppose that earthquakes in a given season at a particular place occur following a Poisson Process with parameter $\lambda_1 > 0$ or $\lambda_2 > 0$. It occurs with parameter λ_1 for $100p\%$ of the time and with parameter λ_2 for $100(1-p)\%$ of the time, where $p \in (0,1)$ is known.

(a) Describe the distribution of the number of earthquakes during the first t units of time, $t > 0$.

(b) Given that 4 earthquakes occured during the first 10 units of time, find the conditional probability that the parameter value was λ_2.

Hints: The number of earthquakes during $[0, t], t \geq 0$ follows a conditional Poisson process with the rate parameter following the discrete distribution: value λ_1 with probability p and value λ_2 with probability $1 - p$.

Question 4. (3 + 1 + 1 marks) Let $\{N_t\}_{t \geq 0}$ be a Poisson Process with rate $\lambda > 0$. Fix $0 < r < s$ and let n be a positive integer. Find the conditional distribution of N_r given $N_s = n$. Also compute the mean and variance of this conditional distribution.

Hints: Since $N_r \leq N_s$, the conditional distribution $N_r \mid N_s = n$ is supported on $\{0, 1 \cdots, n\}$. For any k in the support,

$$\mathbb{P}(N_r = k | N_s = n) = \frac{\mathbb{P}(N_r = k, N_s = n)}{\mathbb{P}(N_s = n)} = \frac{\mathbb{P}(N_r = k, N_s - N_r = n - k)}{\mathbb{P}(N_s = n)}$$

Use independent increments property to simplify the numerator.